

28 B Using Einstein's Photoelectric Equation,

$$hf = \Phi + eV_s$$

$$3.5 \times 10^{-19} = \Phi + (1.6 \times 10^{-19})(0.25)$$

$$\Phi = 3.1 \times 10^{-19} \text{ J}$$

29 C Total KE before reaction = $7.50(225) = 1782.65 \text{ MeV}$